MAST30027: Modern Applied Statistics

Assignment 3, 2026 — Recovering Lost Reef Classifications

AUTHOR
Student name and ID

Due: 5:00 pm Friday 9 October 2026

Late submissions will not be accepted unless an extension has been granted by the Special Consideration team.

This assignment is worth **11%** of your final mark. There are **33 marks** available across the seven assessed parts.

> **How this assignment is marked**
>
> We will mark a **selected set of parts**. We will select parts worth at least **50%** of the full marks listed. The parts to be marked will be selected after the due date and will not be announced in advance.
>
> You must therefore attempt **every** part. A selected part that has not been attempted scores zero.
>
> The mark allocations shown against each part indicate its relative weight and the depth of working expected, and apply if that part is selected.

Submission and working

Create a report using the supplied QMD template and replace the author placeholder with your name and student ID. Create an RStudio Project for the assignment, keep the QMD and TXT files together in the project folder, and use relative file paths. If you include images of handwritten working, keep them in an `images` folder inside the same project.

Handwritten working

The mathematical derivations in Problems 1 and 2 may be typed using LaTeX or completed clearly by hand and included as a picture in the relevant part of the report. If you use handwritten working, scan or photograph it, crop it closely, and save it as a PNG image. The image must be upright, high contrast, legible at normal magnification, and labelled with the relevant part number.

Store the image in the `images` folder beside your QMD. Include it by placing `knitr::include_graphics()` **inside an R code chunk**, as illustrated below. Replace the example path with the relative path to your own image and remove `#| eval: false` when the file is present.

```
knitr::include_graphics("images/problem1-part-a-working.png")
```

> **Parts that must be typed**
>
> **Problem 1 Part (d)** and **Problem 2 Part (c)** must be typed. Handwritten solutions and screen-captured R code, output, or figures will not be marked for those parts. Your R

code must appear as live code in the rendered document.

Handwritten working must appear in the rendered report and final PDF. It may support mathematical derivations, but it does not replace required R code, output, or written interpretation. Do not submit separate image files.

Final submission

Render the QMD to HTML, open the HTML in a browser, and use **Print** and **Save as PDF** to create `studentID_assignment3.pdf`. Check that all code, output, equations, figures, and handwritten working are readable and are not cut off. Submit this single PDF through the LMS. Do not submit the QMD, HTML, TXT, or separate image files. It is your responsibility to upload the correct PDF on time.

Consulting engagement

Your role and client

You are a **Graduate Statistical Consultant** at **Meridian Insights**, the fictional analytics consultancy you worked for in Assignment 2. The same reef-monitoring program has returned with a new problem, and this time the difficulty is not the model for the counts but the fact that a crucial variable is missing.

In Assignment 2 you analysed coral trout counts at monitoring sites for which the management zone of each reef was recorded. The client could tell you which reefs were open to fishing and which were in no-take zones, so the explanatory variables were known and a GLM was the natural tool.

The client now has the opposite problem.

The client's problem

During a database migration, the reef-monitoring program lost the management-classification table that linked each surveyed reef to its level of fishing pressure. The survey counts themselves were recovered intact, but for a large block of historical surveys **the classification of each reef is simply gone**.

The Monitoring Manager writes:

> "We have 300 reef-season counts from the recovered archive and no idea which management class each reef belonged to. From the program's design we are confident there were three classes in use, with quite different levels of fishing pressure, but the labels are lost and the original field sheets were destroyed.
>
> We do not need the labels back for their own sake. What we need to know is: how many of these reefs fell into each class, what the typical count was in each class, and — for any individual reef — how confident we can be about which class it came from. Some of our reporting obligations depend on that split.
>
> Is it possible to recover any of this from the counts alone?"

This is a question about **latent structure**. There is a grouping variable that genuinely exists and genuinely influences the counts, but it was never observed for these reefs. A finite mixture model is designed for exactly this situation: it treats the missing class label as an unobserved random variable, and the EM algorithm lets us obtain maximum likelihood estimates of the parameters despite never seeing it.

Scientific background

Coral trout in the genus *Plectropomus* are conspicuous reef fishes and important fishery species on the Great Barrier Reef. Standardised underwater visual censuses along fixed transects are the usual way reef fish abundance is monitored, as described on the AIMS pages for fish research and monitoring and the Long-Term Monitoring Program.

Fishing pressure is one of the stronger influences on coral trout abundance, so it is plausible that reefs under three distinct management regimes would show three distinct typical abundances. That biological expectation is what makes a three-component mixture a reasonable starting model here — but note that the expectation comes from outside the data, and part of your job is to check that the fitted model is consistent with what you observe.

> ⚠ **These data are simulated**
>
> The counts supplied with this assignment are **simulated for teaching purposes**. They are not AIMS records, and the "lost classification table" is a constructed scenario. Do not present any conclusion from this assignment as a finding about the real Great Barrier Reef. The scenario is realistic, and the statistical problem is a real one, but the numbers are not.

The data

Each observation is the **total number of coral trout counted at one reef over one monitoring season**, summed across all standardised transects surveyed at that reef in that season. Counts are non-negative integers, the reefs were surveyed independently of one another, and survey effort has already been standardised, so the counts are directly comparable.

Two files are supplied.

File	Contents
`assignment3_prob1_2026.txt`	300 reef-season counts from the recovered archive. The management class of each reef is **unknown**.
`assignment3_prob2_2026.txt`	100 further reef-season counts, recovered later from a separate backup in which the metadata survived. Every reef in this file is **known to belong to the second**

class.

Keep both files in the same folder as this QMD file. Read them using:

```
X <- scan(file = "assignment3_prob1_2026.txt", what = double())
length(X)
hist(X)
```

Remove `#| eval: false` once the data files are correctly located and you are ready to begin.

Problem 1: Recovering the three classes (18 marks)

We model the 300 recovered counts as a mixture of three Poisson distributions. Write $X_1, \ldots, X_{300}$ for the counts and assume they are independent of one another. For each i, let Z_i denote the unobserved management class of reef i, and assume

$$
\begin{aligned}
Z_i &\sim \text{categorical}(\pi_1, \pi_2, 1 - \pi_1 - \pi_2), \\
X_i \mid Z_i = 1 &\sim \text{Poisson}(\lambda_1), \\
X_i \mid Z_i = 2 &\sim \text{Poisson}(\lambda_2), \\
X_i \mid Z_i = 3 &\sim \text{Poisson}(\lambda_3),
\end{aligned}
$$

where the Poisson distribution has probability mass function

$$
f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \ldots
$$

Here π_k is the proportion of reefs in class k and λ_k is the expected seasonal count for a reef in class k. We aim to obtain the MLE of $\theta = (\pi_1, \pi_2, \lambda_1, \lambda_2, \lambda_3)$ using the EM algorithm.

Part (a): Expected complete log-likelihood (5 marks)

Let $X = (X_1, \ldots, X_{300})$ and $Z = (Z_1, \ldots, Z_{300})$. Derive the expectation of the complete log-likelihood,

$$
Q(\theta, \theta^0) = E_{Z|X,\theta^0}\left[\log P(X, Z \mid \theta)\right].
$$

Show your working. Express your final result in terms of the responsibilities (i.e., the posterior membership probabilities $P(Z_i = k \mid X_i, \theta^0)$).

Part (b): E-step (3 marks)

Derive the E-step of the EM algorithm. Give an explicit expression for $P(Z_i = k \mid X_i, \theta^0)$ in terms of θ^0 and x_i.

Part (c): M-step (5 marks)

Derive the M-step of the EM algorithm. Obtain the updates for π_1, π_2 and for $\lambda_1, \lambda_2, \lambda_3$, showing how the constraint $\pi_1 + \pi_2 + \pi_3 = 1$ is handled.

Part (d): Implementation (5 marks)

> ⓘ **Must be typed**

Implement the EM algorithm and obtain the MLE of the parameters by applying your implementation to $X_1, \ldots, X_{300}$. Stop the EM iterations when either the number of iterations reaches 100 (`max.iter = 100`) or the incomplete log-likelihood has changed by less than 0.00001 ($\epsilon = 0.00001$).

Run the EM algorithm twice, from the two sets of initial values below, and report the estimates from the run with the higher incomplete log-likelihood.

	π_1	π_2	λ_1	λ_2	λ_3
1st initial values	0.3	0.3	3	20	35
2nd initial values	0.1	0.2	5	25	40

For each run, check that the incomplete log-likelihoods increase at each EM-step by plotting them.

Report your final estimates of $\pi_1, \pi_2, \pi_3, \lambda_1, \lambda_2, \lambda_3$ to a sensible precision.

Problem 2: Using the recovered labelled surveys (15 marks)

A second batch of counts has since been recovered from a separate backup. For these 100 reef-seasons the metadata survived, and all of them are known to come from the **second** class. The client wants to know whether using them alongside the unlabelled archive improves the picture.

Read both files as follows:

```
X  <- scan(file = "assignment3_prob1_2026.txt", what = double())
X0 <- scan(file = "assignment3_prob2_2026.txt", what = double())
length(X)
length(X0)
par(mfrow = c(1, 2))
hist(X0)
hist(c(X, X0))
```

Let $X_1, \ldots, X_{300}$ denote the counts from `assignment3_prob1_2026.txt` and $X_{301}, \ldots, X_{400}$ the counts from `assignment3_prob2_2026.txt`. Assume $X_1, \ldots, X_{400}$ are independent of one another. We model $X_1, \ldots, X_{300}$ using the same three-component mixture as in Problem 1, so that for $i = 1, \ldots, 300$,

$$Z_i \sim \text{categorical}(\pi_1, \pi_2, 1 - \pi_1 - \pi_2),$$

$$X_i \mid Z_i = 1 \sim \text{Poisson}(\lambda_1), \quad X_i \mid Z_i = 2 \sim \text{Poisson}(\lambda_2), \quad X_i \mid Z_i = 3 \sim \text{Poisson}(\lambda_3),$$

but the class of the second batch is known, so for $i = 301, \ldots, 400$,

$$X_i \sim \text{Poisson}(\lambda_2).$$

Note that the second batch shares the parameter λ_2 with the mixture but contributes no unobserved Z_i. We again aim to obtain the MLE of $\theta = (\pi_1, \pi_2, \lambda_1, \lambda_2, \lambda_3)$ using the EM algorithm.

Part (a): Expected complete log-likelihood (5 marks)

Let $X = (X_1, \ldots, X_{400})$ and $Z = (Z_1, \ldots, Z_{300})$. Derive the expectation of the complete log-likelihood,

$$Q(\theta, \theta^0) = E_{Z|X,\theta^0}\left[\log P(X, Z \mid \theta)\right].$$

Show your working.

Part (b): E-step and M-step (5 marks)

Derive the E-step and the M-step of the EM algorithm for this model.

Part (c): Implementation (5 marks)

> ⓘ **Must be typed**

Implement the EM algorithm and obtain the MLE of the parameters by applying your implementation to $X_1, \ldots, X_{400}$. Use the same stopping rule as in Problem 1: stop when either the number of iterations reaches 100 (`max.iter = 100`) or the incomplete log-likelihood has changed by less than 0.00001 ($\epsilon = 0.00001$).

Run the EM algorithm twice, from the same two sets of initial values, and report the estimates from the run with the higher incomplete log-likelihood.

	π_1	π_2	λ_1	λ_2	λ_3
1st initial values	0.3	0.3	3	20	35
2nd initial values	0.1	0.2	5	25	40

For each run, check that the incomplete log-likelihoods increase at each EM-step by plotting them.

Note that the incomplete log-likelihood for this model is not the same expression as in Problem 1. State the expression you used.

Presentation requirements

- Create your report using this QMD template and write beneath the supplied headings. You may add subheadings, but do not delete the assignment brief.
- Replace the author placeholder in the YAML with your name and student ID.
- Answers must be clearly numbered and in the same order as the assignment

questions. Because the parts to be marked are not announced in advance, a missing or unlabelled answer cannot be given credit.

- To get full marks, show your working: the R commands and output you used, the mathematical derivation, and a rigorous explanation of why you reach your conclusions. Final answers with no working receive zero marks.
- All R code and results used in the analysis must be visible. Do not paste screenshots of code, console output, tables, or figures.
- The QMD must render successfully from beginning to end in a clean R session and must not rely on objects created interactively.
- Include every package-loading command required by your report. Installation commands such as `install.packages()` must not appear in the submitted report.
- Write your EM implementations yourself using base R. Do not use a package that fits mixture models for you. You may use such a package only as a check, and if you do, say so.
- Report numerical results to a sensible precision. Give every figure an informative caption, label axes clearly, and refer to each figure in the text.
- Cite any external sources, packages, or code you use.

Problem 1

Part (a): Expected complete log-likelihood

Derive $Q(\theta, \theta^0)$ for the three-component Poisson mixture. Show the complete-data likelihood, take logarithms, and take the expectation over $Z \mid X, \theta^0$. Identify the responsibility term.

Part (b): E-step

Give the explicit expression for $P(Z_i = k \mid X_i, \theta^0)$ and explain what it means for one reef.

Part (c): M-step

Maximise $Q(\theta, \theta^0)$ over θ. Handle the constraint on the mixture proportions explicitly, and comment on the form of the update for λ_k.

Part (d): Implementation

Write your EM implementation here as live R code. Run it from both sets of initial values, plot the incomplete log-likelihood against iteration for each run, and report the estimates from the better run together with the number of iterations used.

Problem 2

Part (a): Expected complete log-likelihood

Derive $Q(\theta, \theta^0)$ for the combined model, being explicit about the contribution of the 100 labelled observations.

Part (b): E-step and M-step

Derive both steps. State which updates are unchanged from Problem 1 and why, and set out the update for λ_2.

Part (c): Implementation

Write your EM implementation for the combined model here as live R code. State the incomplete log-likelihood expression you used, run from both sets of initial values, plot the log-likelihood against iteration, and report the better run.